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Robust Alignment: Harmonizing Clean Accuracy and Adversarial Robustness in Adversarial Training

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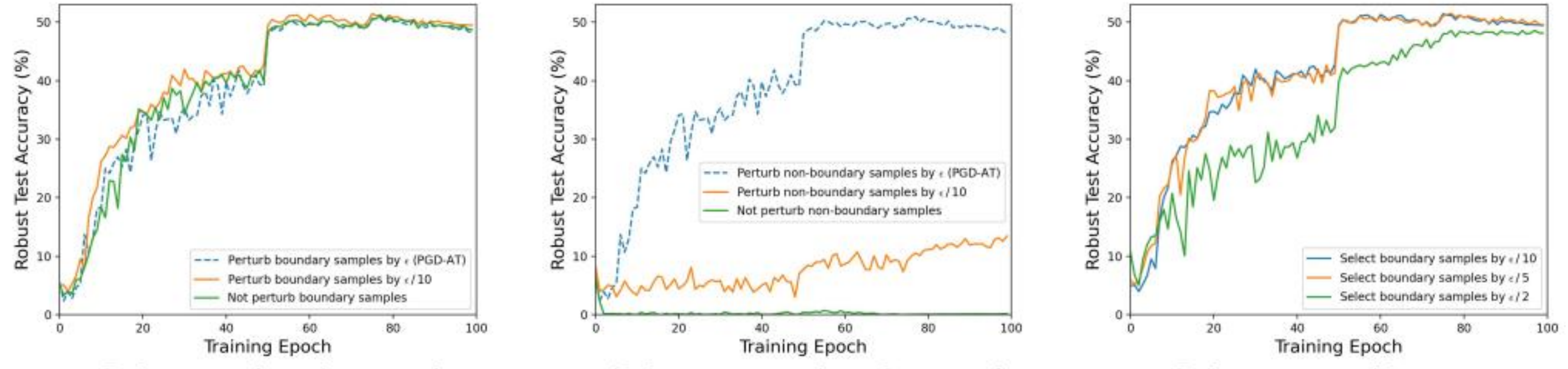
Brief Introduction

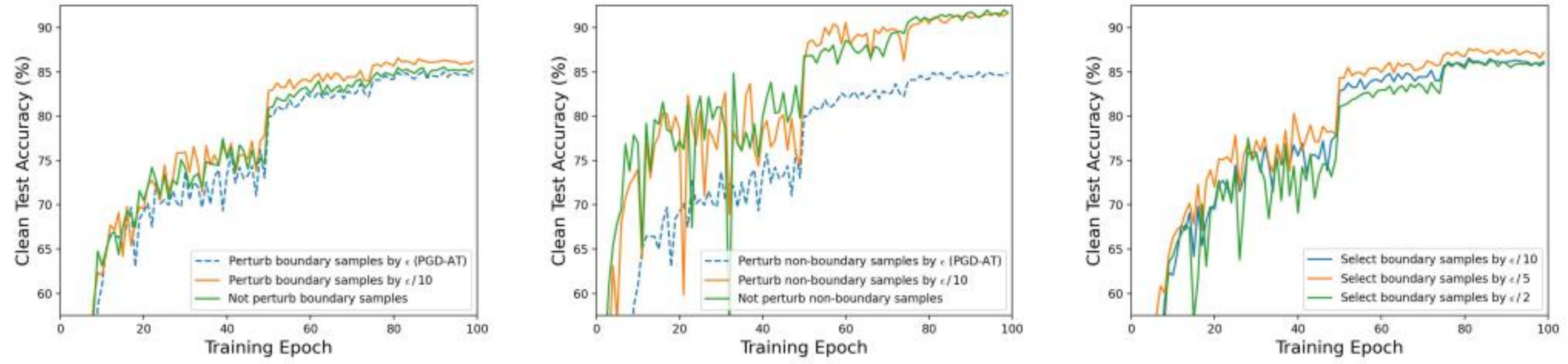
Adversarial Training (AT) faces the ***accuracy-robustness trade-off*** problem. We reveal varying input perturbation intensities for training samples near decision boundaries in AT have ***minimal impact*** on model robustness. This directly exposes the ***inconsistency*** between accuracy and robustness fluctuations, identifying the ***misalignment between input and latent spaces*** as a critical driver of the accuracy-robustness trade-off, which can be then fixed by regularization.

Motivation

(a) and (d): Reducing the intensity of perturbations applied to ***boundary samples*** has basically no influence on the model robustness, while appropriate perturbation does help to improve the clean accuracy. **(b) and (e):** In contrast, fully perturbing ***non-boundary samples*** within the ϵ-ball is vital to learn robustness. **(c) and (f):** The partition of the boundary and non-boundary samples matters and an appropriate strategy can benefit ***both*** robustness and clean accuracy.

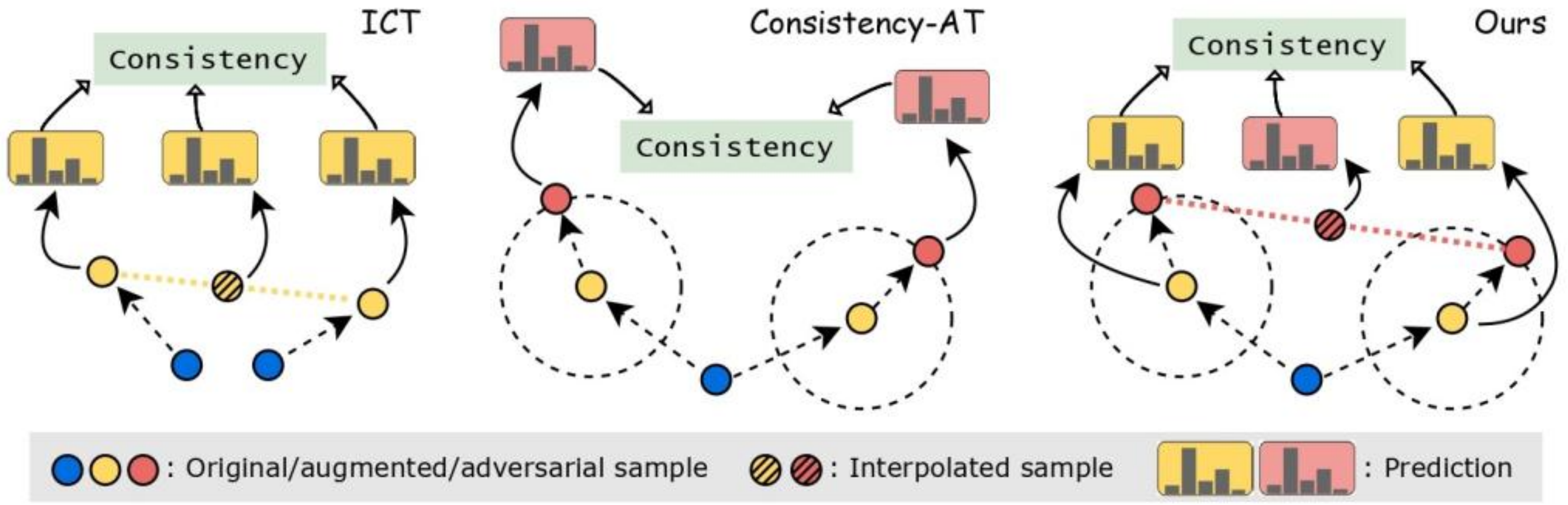


(a) Robust acc - boundary sample (b) Robust acc - non-boundary sample (c) Robust acc - partition strategy



(d) Clean acc - boundary sample (e) Clean acc - non-boundary sample (f) Clean acc - partition strategy

Domain Interpolation Consistency Adversarial Regularization (DICAR)



$$\mathcal{L}^{\mathrm{BCE}}(\mathbf{p}(\hat{\mathbf{x}}_i', \boldsymbol{\theta}), y_i) \cdot \mathbb{1}(h_{\bar{\boldsymbol{\theta}}}(\hat{\mathbf{x}}_i'') = y_i) + \mathcal{L}^{\mathrm{BCE}}(\mathbf{p}(\hat{\mathbf{x}}_i'', \boldsymbol{\theta}), y_i) \cdot \mathbb{1}(h_{\bar{\boldsymbol{\theta}}}(\hat{\mathbf{x}}_i'') \neq y_i)$$
$$+\lambda \cdot \left(\mathcal{L}^{\mathrm{JS}}(\mathbf{p}((\beta \cdot \hat{\dot{\mathbf{x}}}_i' + (1-\beta) \cdot \hat{\ddot{\mathbf{x}}}_i'), \boldsymbol{\theta}) \,||\, (\beta \cdot \mathbf{p}(\dot{\mathbf{x}}_i, \boldsymbol{\theta}) + (1-\beta) \cdot \mathbf{p}(\ddot{\mathbf{x}}_i, \boldsymbol{\theta}))) \cdot \mathbb{1}(h_{\bar{\boldsymbol{\theta}}}(\mathbf{x}_i) = y_i) + \mathcal{L}^{\mathrm{KL}}(\mathbf{p}(\mathbf{x}_i, \boldsymbol{\theta}) \,||\, \mathbf{p}(\hat{\mathbf{x}}_i', \boldsymbol{\theta})) \cdot (1 - \mathbf{p}_{y_i}(\mathbf{x}_i, \boldsymbol{\theta}))\right)$$

Evaluation (3 models, 3 datasets, 4 baselines + 14 SOTAs involved)

Table 2. Comparison of RAAT with **nine related works** on the trade-off with CIFAR-100 and ResNet-18 under ℓ_∞ threat model. Despite the related works usually employ recognized tricks (as in Section 5.3) while our RAAT builds only upon simple PGD-AT, it still achieves the best trade-off.

Method		Clean	AA	Mean
MMA	ICLR'20	60.60 ± 0.60	18.40 ± 0.20	39.50
AWP	NeurIPS'20	55.16 ± 0.27	25.16 ± 0.39	40.16
GAIRAT	ICLR'21	58.43 ± 0.28	17.54 ± 0.33	37.99
MAIL	NeurIPS'21	60.74 ± 0.15	22.44 ± 0.53	41.59
TE	ICLR'22	56.45 ± –	26.30 ± –	41.38
HAT	ICLR'22	59.19 ± 0.07	23.75 ± 0.14	41.47
SOVR	ICML'23	52.10 ± 0.80	24.30 ± 0.20	38.20
ADR	ICLR'24	56.54 ± –	26.42 ± –	41.48
PIAT	TIFS'25	56.04 ± –	26.09 ± –	41.07
RAAT	**Ours**	58.53 ± 0.14	25.65 ± 0.26	**42.09**

Table 3. Comparison of RAAT$^{\#}$ with **11 current SOTA AT methods** either in general or on the accuracy-robustness trade-off problem with PreActResNet-18 under ℓ_∞ threat model. The best results are marked in bold.

Method		CIFAR-10				CIFAR-100			
		Clean	AA	Mean	NRR	Clean	AA	Mean	NRR
WA	UAI'18	83.50	49.89	66.695	62.461	57.26	25.83	41.545	35.601
MMA	ICLR'20	**85.50**	37.20	61.350	51.844	**60.60**	18.40	39.500	28.229
AWP	NeurIPS'20	81.11	50.09	65.600	61.933	54.10	25.16	39.630	34.347
GAIRAT	ICLR'21	78.70	37.70	58.200	50.979	52.00	19.80	35.900	28.680
KD+SWA	ICLR'21	84.06	49.82	66.940	62.562	57.17	25.66	41.415	35.422
EWAT	ICML'21	82.80	48.20	65.500	60.931	54.20	23.52	38.860	32.805
MAIL	NeurIPS'21	79.50	39.60	59.550	52.867	46.50	16.70	31.600	24.574
TE	ICLR'22	82.04	50.12	66.080	62.225	56.41	25.84	41.125	35.444
SOVR	ICML'23	81.90	49.40	65.650	61.628	52.10	24.30	38.200	33.142
ReBAT	NeurIPS'23	82.09	50.72	66.405	62.700	56.13	27.60	41.865	37.004
RPAT	ICCV'25	82.27	50.75	66.510	62.776	56.43	27.63	42.030	37.096
RAAT$^{\#}$	**Ours**	82.76	**51.65**	**67.205**	**63.605**	56.92	**28.08**	**42.500**	**37.607**

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[Code] https://github.com/FlaAI/RAAT
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